



香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

PHY1001 Mechanics

Lecture 17: Gravitation

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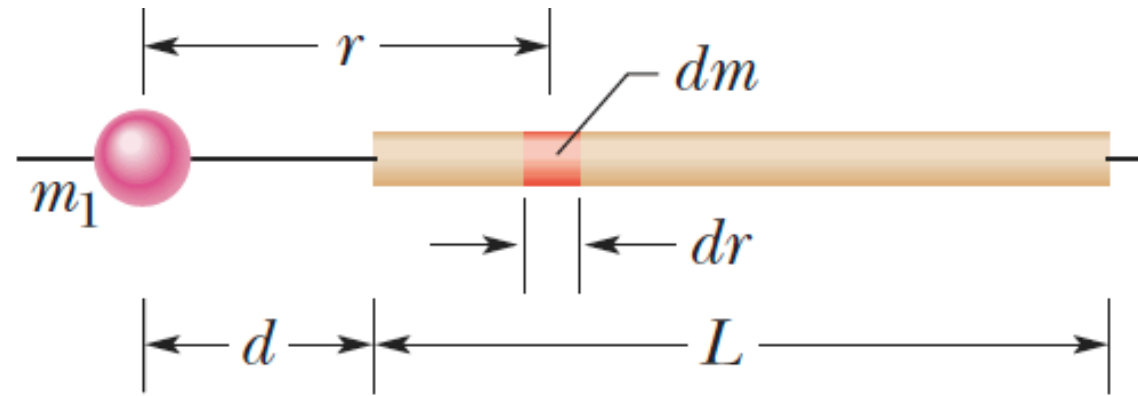
Recap: Elasticity

		Stress	Strain	Elastic modulus
		F/A	Fractional deformation	Stress/strain
Tensile	Solid	$\frac{F_{\perp}}{A}$	$\frac{\Delta l}{l_0}$	$Y = \frac{F_{\perp} l_0}{A \Delta l}$ (Young's modulus)
Shear		$\frac{F_{\parallel}}{A}$	$\frac{x}{h} (x \ll h)$	$S = \frac{F_{\parallel} h}{Ax}$ (Shear modulus)
Bulk	Fluid (liquid, gas)	ΔP	$\frac{\Delta V}{V}$	$B = -\frac{\Delta P}{\Delta V/V}$ (Bulk modulus)

Newton's Law of Gravity

- $\vec{F} = -G \frac{Mn}{r^2} \hat{r}$, where
 - G : gravitational constant, $6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$
 - $\hat{r}$: unit vector of $\vec{r}$

Gravity from A Rod



A particle of mass m_1 is at a distance d from one end of a uniform rod with length L and mass M . Find the magnitude of gravitational force on the particle from the rod.

$$dF = G \frac{m_1 dm}{r^2}$$

$$dm = \frac{dr}{L} M$$

$$dF = G \frac{m_1 dr M}{r^2 L}$$

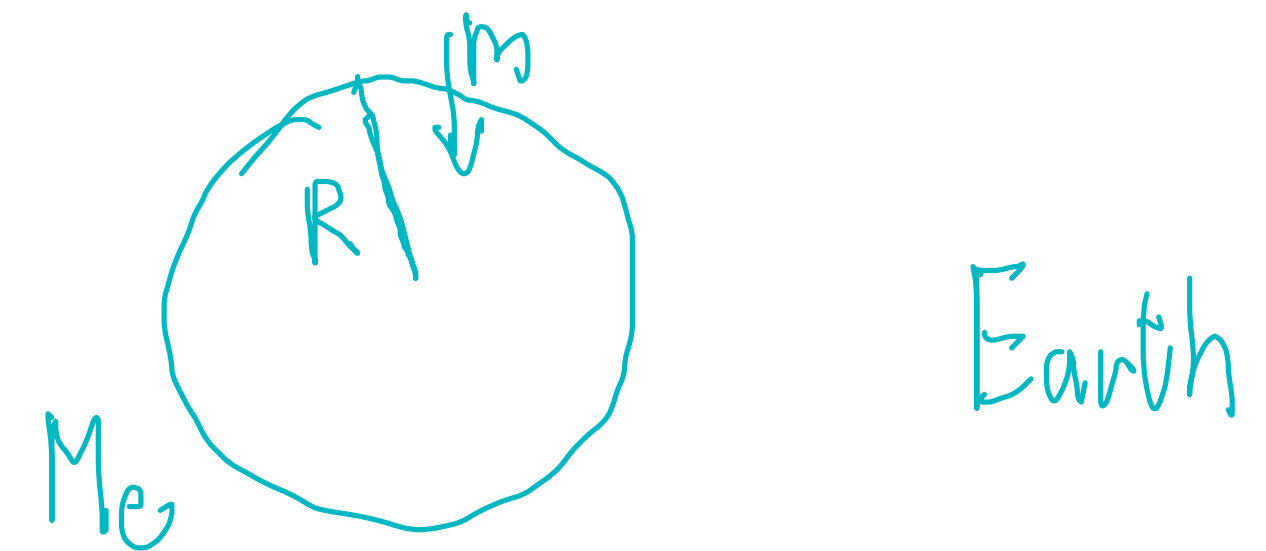
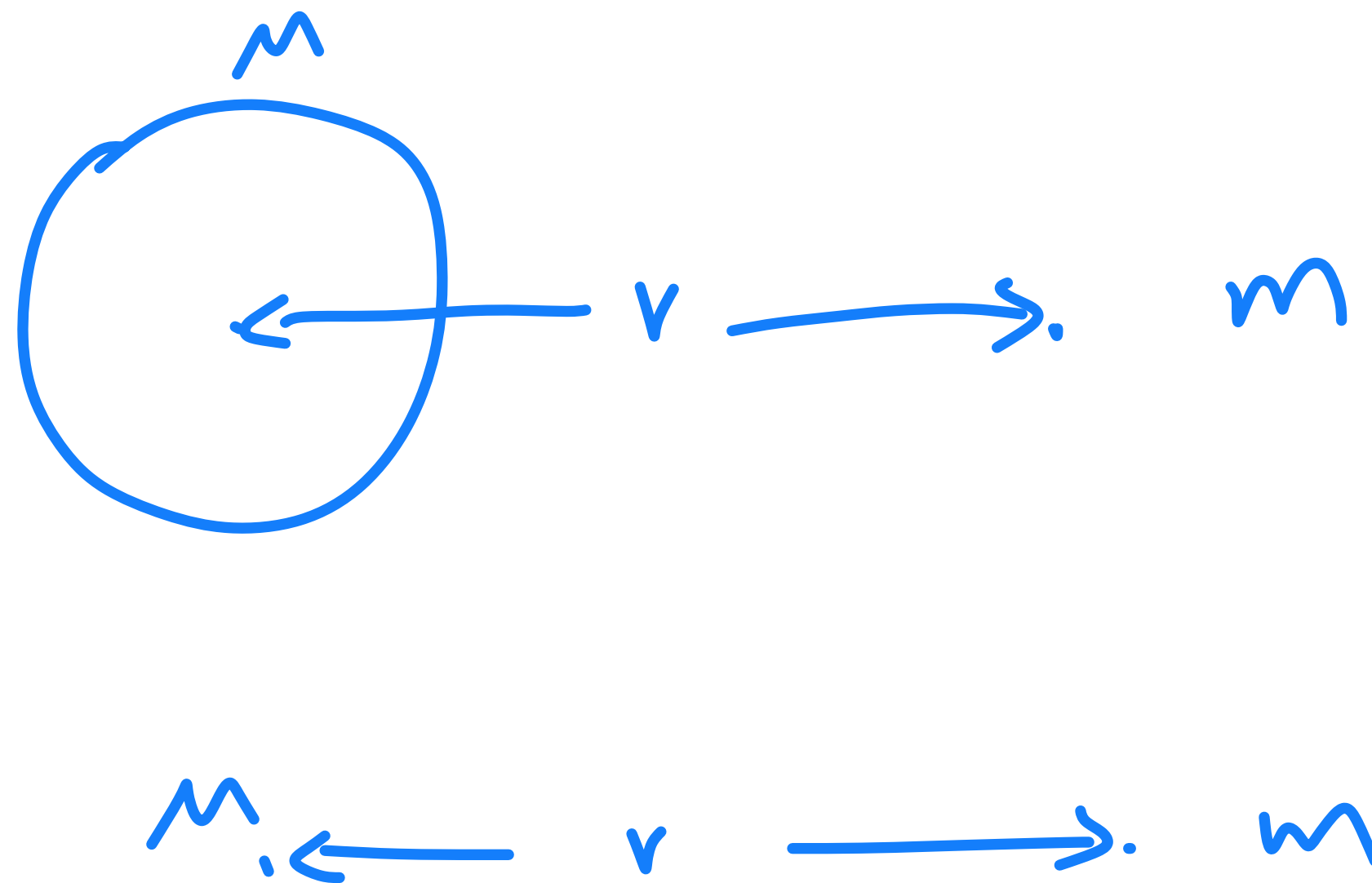
$$= G \frac{M m_1}{L} \cdot \frac{dr}{r^2}$$

$$F = \int dF = \int_d^{d+L} G \frac{M m_1}{L} \cdot \frac{dr}{r^2}$$

$$= G \frac{M m_1}{d(d+L)}$$

Gravity Outside a Uniform Shell/Sphere

- The gravity from a uniform shell/sphere on an object outside the shell/sphere is equivalent to the gravity from a point at the center of the shell/sphere with the same mass



$$mg = G \frac{M_E m}{R^2}$$

Weight

The gravitational acceleration constant $g = 9.8 \text{ m/s}^2$ is due to the gravity imposed by the Earth. The Earth can be viewed as a solid sphere with radius $R = 6.4 \times 10^6 \text{ m}$. Can you find the mass of the Earth M_e ?

$$g = G \frac{M_e}{R^2}$$

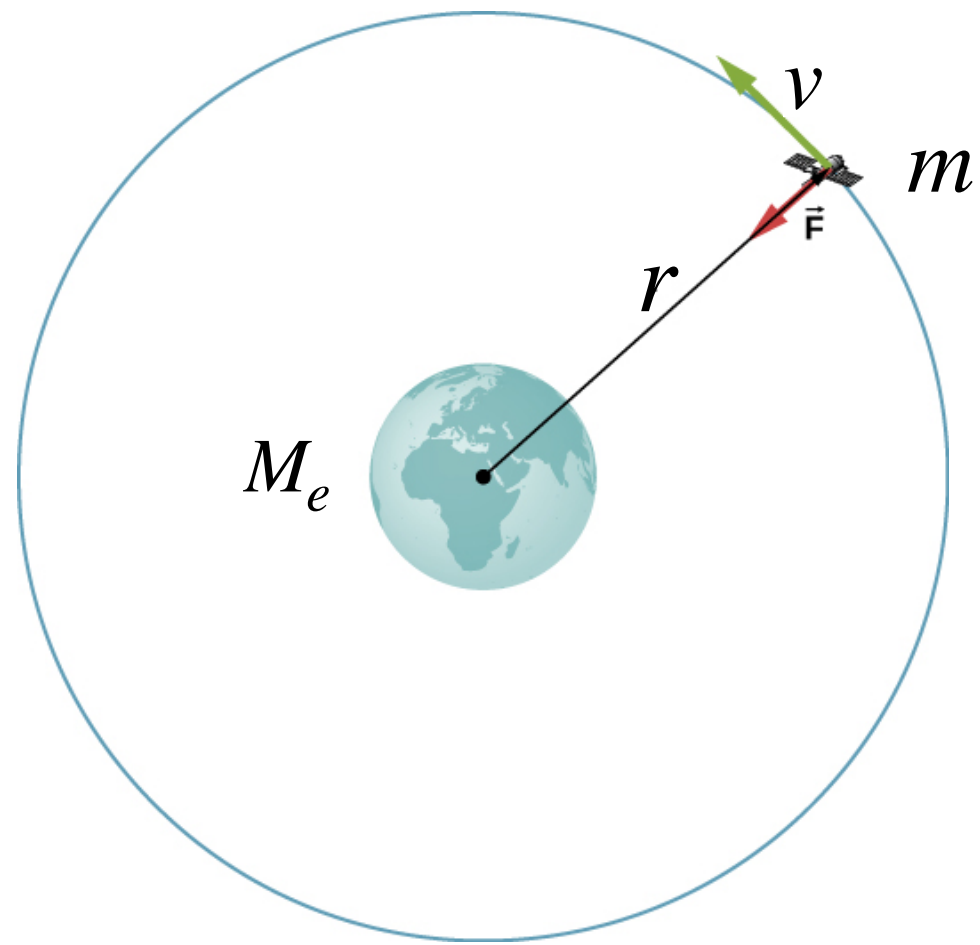
$$M_e = \frac{gR^2}{G}$$

$$= 6.02 \times 10^{24} \text{ kg}$$

$$1 \text{ mile} \approx 1.6 \text{ km}$$

$$R \approx 4000 \text{ mile}$$

Motion of Satellites



What are the angular velocity ω , velocity v , and period T of a satellite?

$$G \frac{M_e m}{r^2} = m \omega^2 r$$
$$\omega = \sqrt{\frac{G M_e}{r^3}}$$

$$T = \frac{2\pi}{\omega}$$
$$= 2\pi \sqrt{\frac{r^3}{G M_e}}$$

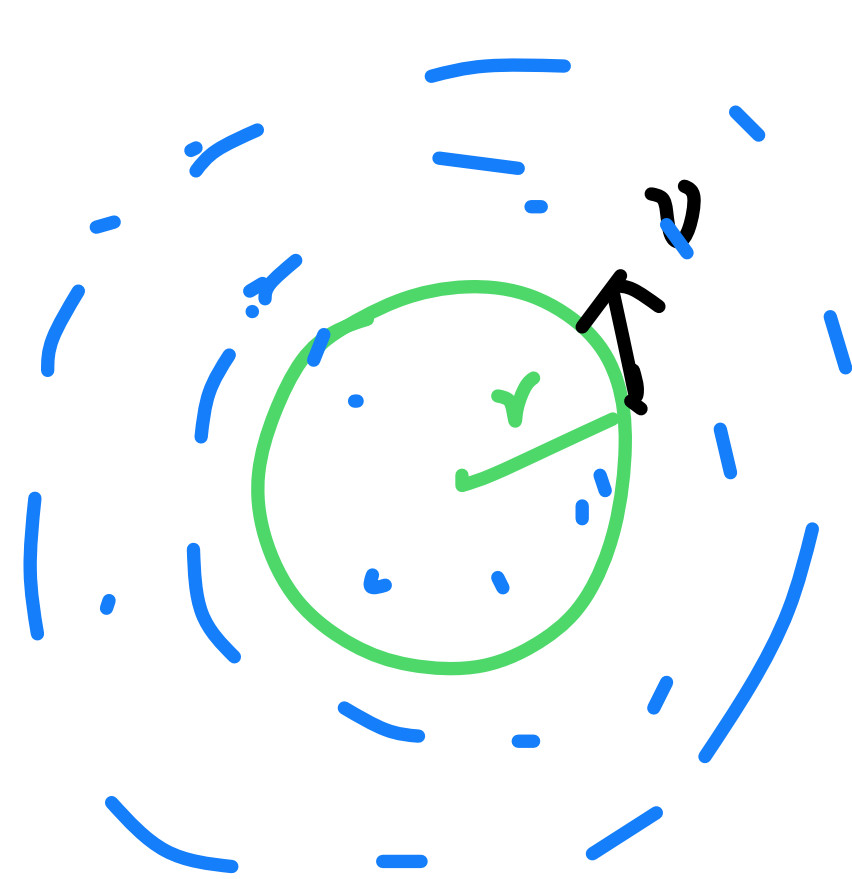
$$G \frac{M_e m}{r^2} = m \frac{v^2}{r}$$
$$v = \sqrt{\frac{G M_e}{r}}$$
$$(v = \omega r)$$

Gravity in A Uniform Shell

- A uniform shell of matter exerts no net gravitational force on a particle located inside it.

$$F_g = G \frac{M_e \frac{4\pi r^3}{3} \rho}{r^2}$$
$$= G \frac{M_e m}{R^3} \cdot r$$

Galaxy Rotation Curve

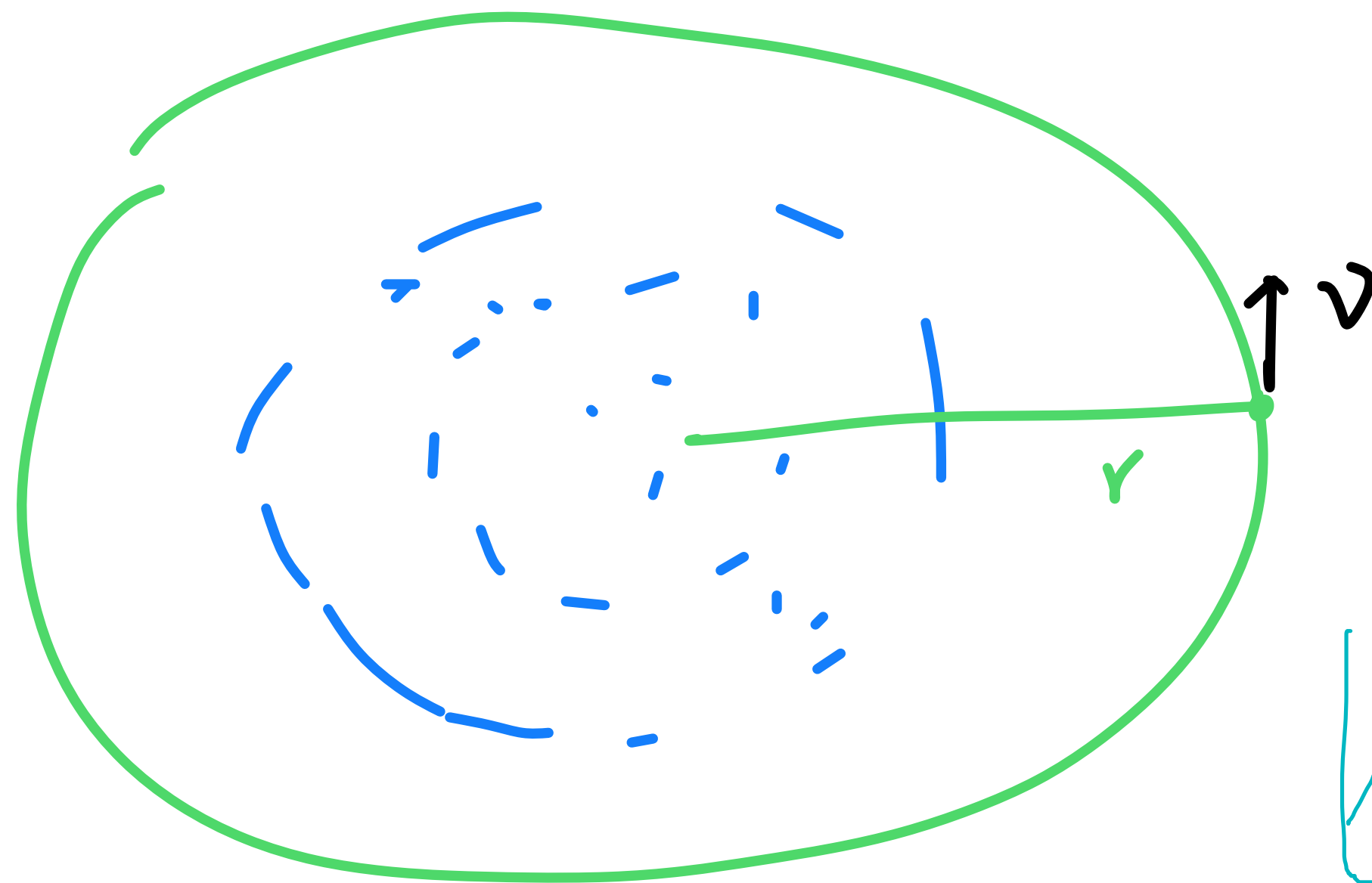


$$R_0 > R$$

$$F_g = G \cdot \rho \cdot \frac{\frac{4}{3}\pi R^3 \cdot m_s}{R^2} = m_s \frac{v^2}{R}$$

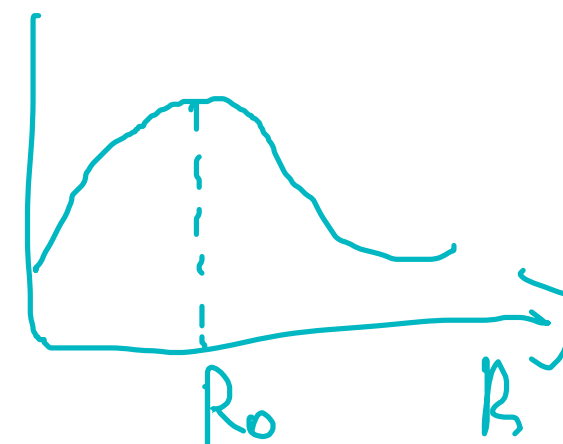
$$R < R_0 \quad v = \sqrt{G \rho \frac{4}{3}\pi} \cdot R$$

$$R > R_0$$

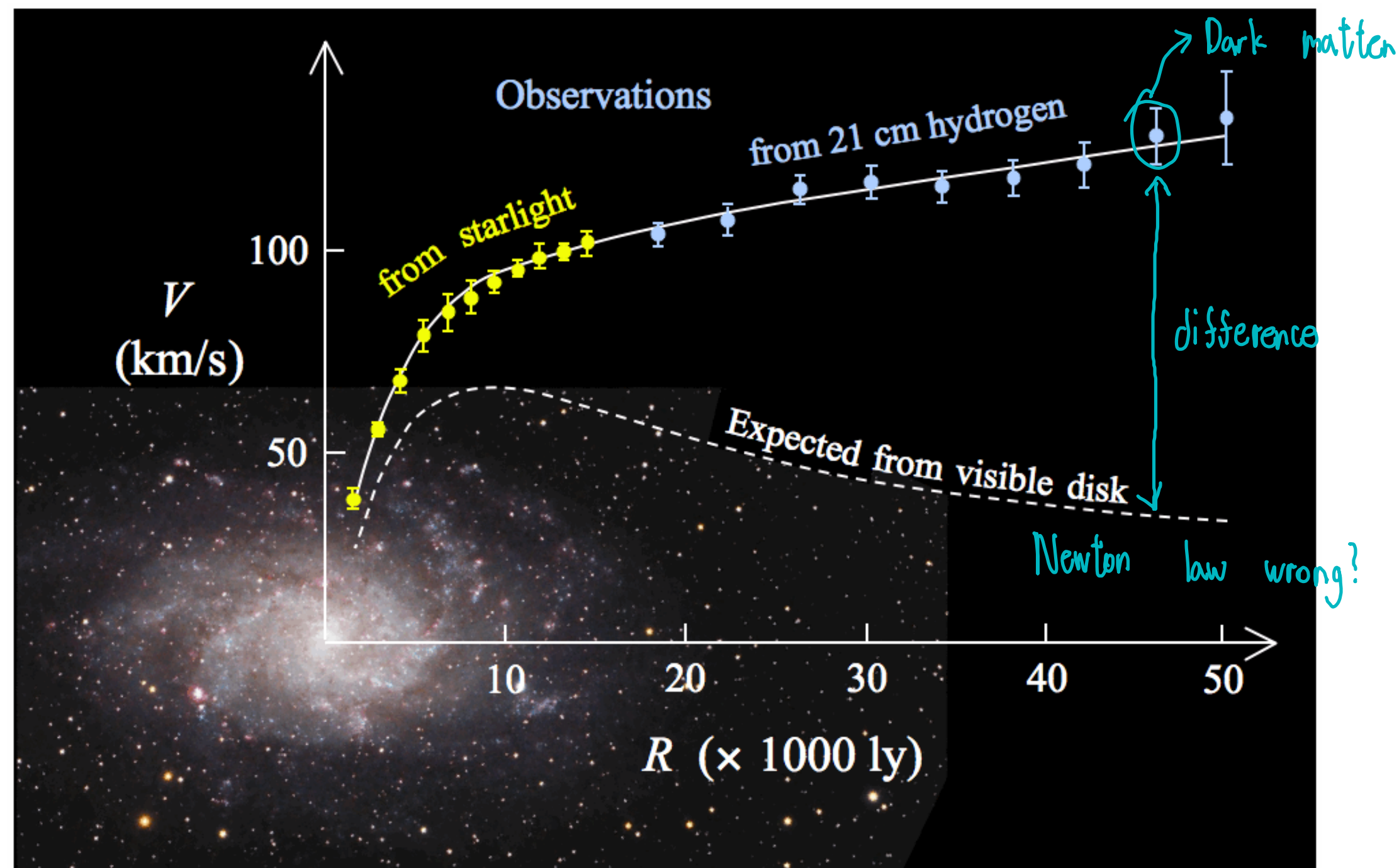


$$F_g = G \frac{M_G \cdot m_s}{R^2} = m_s \frac{v^2}{R}$$

$$v = \sqrt{\frac{G M_s}{R}} = \sqrt{G M_G} \cdot \frac{1}{\sqrt{R}}$$



Evidence of Dark Matter



Rotation curve



Vera Rubin (1928 - 2016)

Gravitational Potential Energy

- Previously: $U_g = mgh$
- $U = -G \frac{Mm}{r}$

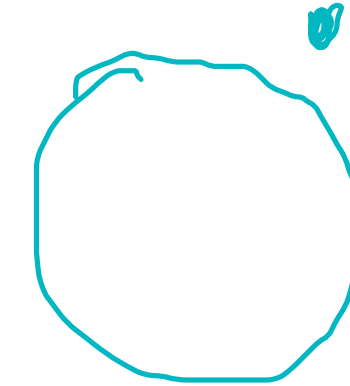
$$mg = G \frac{M_m}{R^2}$$

too large

$$\begin{aligned} dU &= -dW \\ &= -F ds \end{aligned}$$

$$\Delta U = \int_{\infty}^r G \frac{Mm}{r^2} dr$$

$$U_r - \underset{\substack{\uparrow \\ 0}}{U_{\infty}} = -G \frac{Mm}{r} = U_g(r)$$



Escape Velocity

- $K + U > 0$ at the surface of the Earth
- $V_{\text{esc}} = \sqrt{\frac{2GM_e}{R}}$, the velocity to escape Earth

$$E = K + U_g \geq 0$$

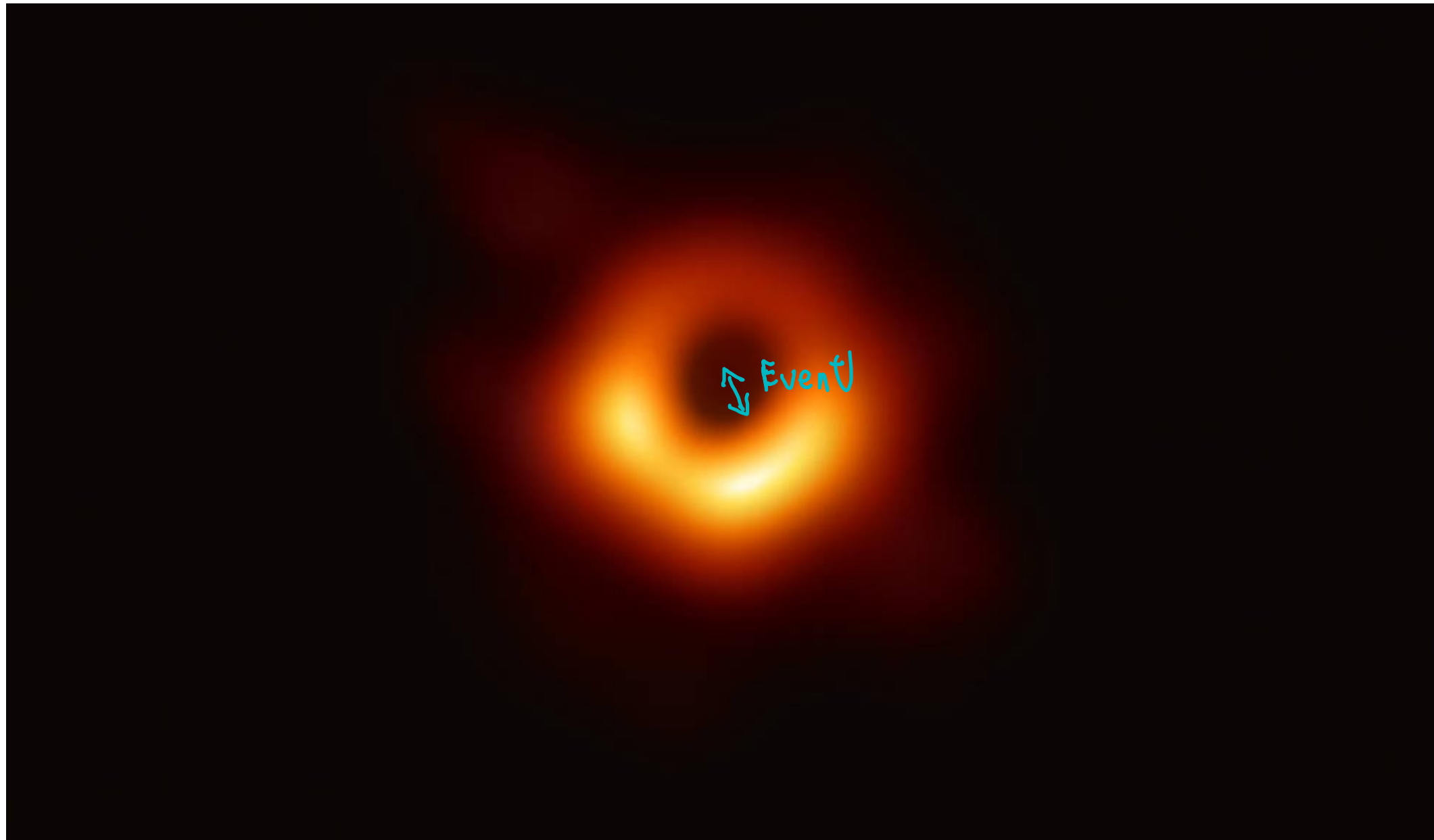
$\downarrow \qquad \downarrow$
 $\geq 0 \qquad 0$

$$E_i = K_i + U_i \geq 0$$

$$\frac{1}{2}mv^2 + \left(-G \frac{M_e m}{R}\right) > 0$$

$$v > \sqrt{\frac{2GM_e}{R}} \approx 11.2 \text{ km/s}$$

Black Holes



Event Horizon Telescope (EHT) Collaboration, 2019

- Event horizon: the boundary of black holes, within which light cannot escape.
- Radius of event horizon: $R = \frac{2GM}{c^2}$ ← speed of light
- If a black hole has the mass of Earth, its event horizon is less than 1 cm!

$$v_e = \sqrt{\frac{2GM}{R}} = c$$

$$R = \frac{2GM}{c^2}$$